Modern Algebra 2

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MA 542

Abstract

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Contents

Chapter 1

Rings

Lecture 1: Syllabus Day

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Definition 1.1: Ring

Let R be a set together with two binary operations

$$+, \bullet : R \times R \to R$$

such that $\forall a, b, c \in R$,

- $a + b = b + a$;
- $(a + b) + c = a + (b + c)$;
- $\exists 0 \in R$ such that $a + 0 = a$ for all $a \in R$;
- $\forall a \in R$, $\exists b \in R$ such that $a + b = 0$;
- $(ab)c = a(bc)$;
- $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

We then call $(R, +, \cdot)$ a ring.

Note that $(R, +)$ is an abelian group. There then clearly exists a forgetful functor **Ring** $\to$ **Grp** that maps $(R, +, \cdot) \mapsto (R, +)$. Examples of rings include $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ under the standard definitions of addition and multiplication. However, these rings are all commutative, and many are in fact fields. There is more to ring theory than commutative rings. Oh also $\mathbb{Z}_n$ is a ring $\forall n \in \mathbb{N}$.

Note that since left multiplication is a faithful group action on itself, every row/column of the cayley table in some group has each number appear exactly once.

We take the convention that rings need not have 1. Homomorphisms thus need not preserve 1, except for epimorphisms. An example of a ring without 1 is $2\mathbb{Z}$. Also note that rings need not be commutative; for example, $M_2(\mathbb{R})$. We generally divide rings into commutative rings and noncommutative rings. the category of commutative rings is denoted **CRing**.

Definition 1.2: Unity

Let R be a ring with some $1 \in R$ such that $\forall r \in R$, $1r = r1 = r$. We say 1 is the unity element of R.

Definition 1.3: Polynomial Ring

Let R be a ring. Define $R[x]$ as the polynomial ring

$$R[x] \coloneqq \left\{ \sum_i a_i x^i \mid a_i \in R \right\}$$

of *finite* linear combinations in the indeterminant x.

Lecture 2: Basic Ring Properties

Fri 24 Jan 2025 13:27

Let R be a ring without unity. Define $S = R \times \mathbb{Z}$ or $S = r \times \mathbb{Z}_{\text{char}(R)}$. Define multiplication by

$$(r, n)(s, m) = (rs + ns + mr, nm).$$

Then $(0, 1)$ is a unity and there exists an inclusion $R \hookrightarrow S$ which is also a ring homomorphism by the map $r \mapsto (r, 0)$.

The quaternions $\mathbb{H}$ are defined as the set of linear combinations of the set $\{1, i, j, k\}$ wherein multiplication satisfies $ijk = -1$, $i^2 = -1$, $j^2 = -1$, and $k^2 = -1$. They are intended as an extension of the complex numbers $\mathbb{C}$, in a similar way to how the complex numbers extend $\mathbb{R}$. However, this definition of multiplication necessitates $ij = -ji$, meaning $\mathbb{H}$ is a noncommuative ring. Furthermore, it can be shown (with some difficulty) that all elements of $\mathbb{H}$ have multiplicative inverses. So $\mathbb{H}$ is a noncommutative ring with unity with multiplicative inverses. We call this a divison ring.

Definition 1.4: Division Ring

Let R be a ring with unity, such that $\forall r \in R$, $\exists r^{-1} \in R$ such that $rr^{-1} = r^{-1}r = 1_R$. Then R is a division ring.

Going back down in dimension to $\mathbb{C}$ gives more structue. Elements in $\mathbb{C}$ do, in fact, commute under multiplication; this makes $\mathbb{C}$ into a field.

Definition 1.5: Field

Let k be a commutative ring with unity, such that for all $r \in k$, there exists $r^{-1} \in k$ such that $rr^{-1} = 1_k$. Then k is called a field.

Definition 1.6: Direct Product

Let R, S be rings. Define the direct product $R \times S$ as the cartesian product of the sets R, S together with coordinate-wise addition and multiplication. Moreover, for any set of rings $R_1, \ldots, R_n$, define the product $R_1 \times \cdots \times R_n$, which is defined in the same way.

The direct product satisfies the universal property for products in **Ring**, but there are more multiplicative structures that can be endowed to the underlying abelian group $R_1 \oplus \cdots \oplus R_n$, which we may or may not see when we do polynomial rings. We may write $\oplus$ instead of $\times$ in some commutative cases.

Proposition 1.1

Let R be a ring, and let $a, b, c \in R$. Then

- $a0 = 0a = 0$;
- $a(-b) = (-a)b = -ab$;
- $(-a)(-b) = ab$;
- $a(b-c) = ab - ac$ and $(a-b)c = ac - bc$;
- $(-1)a = -a$;
- $(-1)(-1) = 1$.

Proof. (1) Many of these are trivial, so I will only prove a few. Let $a \in R$. Then

$$0a = (0+0)a = 0a + 0a \implies 0a - 0a = 0a - 0a - 0a \implies 0 = 0a.$$

(3) Now let $a, b \in R$ and note that $(-a+a)(-b) = 0(-b) = 0$, and thus

$$(-a)(-b) + a(-b) = 0.$$

By property 2,

$$(-a)(-b) + a(-b) = (-a)(-b) - (ab).$$

Therefore, $(-a)(-b) = ab$.

Statement holds for $n = 1, 2$ trivially. Let n even and suppose the statement holds $\forall m \leq n$. We have

$$x^{n+2} = \left(x^{\frac{n}{2}+1}\right)^2 = 0 \iff x^{\frac{n}{2}+1} = 0.$$

Therefore, $x^{n+2} = 0$ iff $x = 0$. Now we have immediately

$$x\left(x^{n+1}\right) = x^{n+2}.$$

If $x \neq 0$ but $x^{n+1} = 0$, then we have $x^{n+2} = 0$ for $x \neq 0$, contradiction. ■

Lecture 3: Rings with Unity

Mon 27 Jan 2025 13:25

Since every ring can be "given" unity, we generally assume rings have unity, because it makes life much easier.

Definition 1.7: Unit

Let R be a ring with unity, and let $a \in R$ such that there exists $b \in R$ such that $ab = 1$. Then a, b are units, and we say b is the multiplicative inverse of a.

Proposition 1.2

Let R be a ring with unity. Then

- the unity is unique;
- If an element $a \in R$ has a multiplicative inverse, then this inverse is unique.

Proof. Let $1_R, 1_S$ be unities of R. Then $1_R 1_S = 1_R = 1_S$ since they are both unities. Let $a \in R$ be a unit, and let b, c be inverses. Then $1 = ab = ac$, and it follows that

$$bab = bac \implies 1b = 1c \implies b = c.$$

■

Proposition 1.3

Let R be a ring with unity. Then the set $U(R)$ of units of the ring R is a group with respect to ring multiplication.

Proof. Note that $1 \in U(R)$, so clearly $U(R)$ has an identity. We also have that all elements have inverses by definition, and associativity follows from the ring axioms. Let $a, b \in U(R)$. Since $(ab)^{-1} = b^{-1}a^{-1}$, and $a^{-1}, b^{-1} \in U(R)$, we have that $ab \in U(R)$, and $U(R)$ is a group. ■

Theorem 1.1

$$U(\mathbb{Z}_n) = \{r \in \mathbb{Z}_n \mid \gcd(r, n) = 1\}.$$

Proof. If the gcd is not 1, then there exists some multiple $mr = 0$ of r with $m < n$. If r is a unit, then $mrr^{-1} = 0$, so $m = 0$, so $\gcd(r, n) = 1$, which is a contradiction. ■

Cancellation:

Proposition 1.4

Let R be a unital ring and let $a \in R$ be a unit. Then $ab = ac$ implies $b = c$.

Proof. Let $ab = ac$. Then $a^{-1}ab = a^{-1}ac$, and thus $b = c$. ■

An example of a finite field is $\mathbb{Z}_p$ for p prime.

Definition 1.8: Subring

Let R be a ring and let $S \subseteq R$ be a subset. Then S is a subring if it is a ring given the same operations as R; S is a subring if the inclusion map $S \hookrightarrow R$ is a ring homomorphism.

Proposition 1.5

Let $S \subseteq R$, such that for all $a, b \in S$, $a - b \in S$ and $ab \in S$. Then S is a subring of R.

Proof. Trivial. ■

Lecture 4: Domains

Mon 03 Feb 2025 13:22

Observe that

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}.$$

Why not apply this notion of a fraction to domains other than $\mathbb{Z}$?

Proposition 1.6

Let D be a domain. Then there exists a field F called the field of fractions $\mathrm{Frac}(D)$, containing (the inclusion of) D as a subring.

Proof. The construction is as follows. Consider the collection

$$S = \{(a, b) \in D \times D \mid b \neq 0\}.$$

Define an equivalence relation $\equiv$ on S as

$$(a, b) \equiv (c, d) \iff ad = bc.$$

Define F as the set of equivalence classes $D/\equiv$. We can define

$$\frac{a}{b} \coloneqq [(a, b)].$$

The operations $+, \cdot$ are defined as

$$[(a, b)] \cdot [(c, d)] = [(ac, bd)], \qquad [(a, b)] + [(c, d)] = [(ad + bc, bd)].$$

It remains to be shown F is a field. Showing the operations are well-defined is messy and I don't care. Showing everything else is even messier and I care even less. By the method of apathy, F is a field. Finally, note that we can define the inclusion $D \hookrightarrow F$ as $d \mapsto d/1$, and this map is in fact an isomorphism. ■

$\mathrm{Frac}(D)$ satisfies the universal property that it is initial with respect to the property of containing D as a subring. That is, any field containing D must contain $\mathrm{Frac}(D)$.

Definition 1.9: Polynomial Ring

Let R be a commutative ring. Define the polynomial ring $R[x]$ as the set of R-linear combinations in the indeterminate x:

$$R[x] \coloneqq \left\{ \sum_{i=0}^{n} a_i x^i \mid \{x_i\}_{i=0}^{k} \subseteq R \right\}.$$

Equality is defined componentwise and addition and multiplication are defined in the usual way.

Definition 1.10: Degree

Let R be a commutative ring, and let $f(x) \in R[x]$. Then $\deg f$ is the greatest n such that $a_n \neq 0$, where

$$f(x) = \sum_{i=0}^{k} a_i x^i.$$

The degree of the 0 polynomial is $-\infty$ because woooo

Proposition 1.7

If R is an integral domain, then $R[x]$ is an integral domain.

Lecture 5: Polynomial Rings

Wed 05 Feb 2025 13:27

Definition 1.11: Ring Homomorphism

Let $(R, +_R, \bullet_R)$ and $(S, +_S, \bullet_S)$ be rings. A function

$$\phi : R \to S$$

is called a *ring homomorphism* if and only if

$$\phi(a +_R b) = \phi(a) +_S \phi(b), \qquad \phi(a \bullet_R b) = \phi(a) \bullet_S \phi(b).$$

Definition 1.12: Isomorphism

A ring *isomorphism* is a ring homomorphism that is invertible/a bijection. If there exists an isomorphism $R \to S$, then we say R and S are isomorphic, and write $R \cong S$.

Theorem 1.2

Let E be a field with a subfield $F \subseteq E$ and let $\alpha \in E$. Then the evaluation function

$$\phi_\alpha : F[x] \to E$$

is a homomorphism.

Proof. Recall that ϕ_α is the unique extension of the inclusion map $F \hookrightarrow E$ to $F[x]$ that maps $x \mapsto \alpha$. The proof isn't worth writing down. ■

Lecture 6: More on Polynomials

Fri 07 Feb 2025 13:33

I didn't write it down, but we have a division algorithm for polynomials. The following are corollaries.

Corollary 1.1

Let $f(x) \in F[x]$. Then $a \in F$ is a zero of $f(x)$ if and only if $x - a$ is a factor of $f(x)$.

Corollary 1.2

A polynomial of degree n has at most n zeroes, counting multiplicity.

Definition 1.13: Irreducible

A non-constant polynomial $f(x) \in F[x]$ is *irreducible* over F if $f(x)$ cannot be expressed as a product of two lower degree polynomials. If f is not irreducible it is reducable.

Proposition 1.8

Let $f(x) \in F[x]$ for F a field where $\deg(f(x)) \in \{2, 3\}$. Then $f(x)$ is reducible if and only if $f(x)$ has a zero in F.

Proof. Let $f(x) = g(x)h(x)$ with $\deg g(x), \deg h(x) < \deg f(x)$. Then we can assume WLOG $\deg g(x) = 1$ and $\deg h(x) = 1$ or 2. So $g(x) = ax + b$ where $-a^{-1}b$ is a zero of g and thus a zero of f.

Conversely, if $f(c) = 0$ for some $c \in F$, then $x - c$ is a divisor of f and we have $f(x) = (x - c)h(x)$ for some $h(x) \in F[x]$. ■

Definition 1.14: Content

The *content* of an integer polynomial $f(x) \in \mathbb{Z}[x]$ is

$$c(f(x)) = \gcd(a_1, \ldots, a_n).$$

That is, the content is the gcd of the coefficients. A polynomial is primitive if it's content equals 1.

Lemma 1.1 (Gauss' Lemma)

The product of two primitive polynomials is primitive.

Proof. Let $f(x) \in \mathbb{Z}[x]$ and a prime p. If one reduces the coefficients modulo p, then one gets $\overline{f}(x) \in \mathbb{Z}_p[x]$. It's easy to show that

$$h(x) = f(x)g(x) \implies \overline{h}(x) = \overline{f}(x)\overline{g}(x).$$

Suppose $c(f) = c(g) = 1$ and $c(fg) \neq 1$. Then there is some prime p that divides the coefficients of $h = fg$ and thus $\overline{h} = 0$. It follows that either $\overline{f}$ or $\overline{g}$ is 0, since $\mathbb{Z}_p$ is a field, and thus p divides the coefficients. Therefore, their content is not 1, a contradiction. ■

Theorem 1.3

Let $f(x) \in \mathbb{Z}[x]$. If f is reductable over $\mathbb{Q}$ then it is reductable over $\mathbb{Z}$.

Proof. Let $f(x) = g(x)h(x)$ for $g, h \in \mathbb{Q}[x]$. We may assume f is primitive, since otherwise we could divide f and g by $c(f)$. Let a, b be the lcm of the denominators of the coefficients of g and h, respectively. Then

$$abf(x) = ag(x)bh(x).$$

These are now both elements of $\mathbb{Z}[x]$. I don't want to finish writing. ■

Corollary 1.3

Let $f(x) \in \mathbb{Z}[x]$ be monic with a nonzero constant term. If $f(x)$ has a zero in $m \in \mathbb{Q}$, then we may assume $m \in \mathbb{Z}$ with $m|a_0$.

Proof. Let $f(x)$ have $a \in \mathbb{Q}$ a zero. Then f has a linear factor $x - a \in \mathbb{Q}[x]$. By theorem 1.3, f has a factorization with a linear factor $x - m \in \mathbb{Z}[x]$. You then do some witchcraft and find m divides a_0. ■

In the proof of Gauss' lemma, we used the fact that the remainder map $\rho : \mathbb{Z}[x] \to \mathbb{Z}_p[x]$ is a homomorphism. We can use this idea further.

Theorem 1.4 (mod p irreducibility)

Let p be a prime and suppose that $f(x) \in \mathbb{Z}[x]$ with $\deg f \geq 1$. If the reduction $\overline{f}(x) \in \mathbb{Z}_p[x]$ is irreducible over $\mathbb{Z}_p[x]$ and $\deg \overline{f} = \deg f$, then $f(x)$ is irreducible over $\mathbb{Q}$.

Proof. If f is reducible over $\mathbb{Q}$, then it's reducable over $\mathbb{Z}$. Thus, $f = gh$ for $g, h \in \mathbb{Z}[x]$ lower degree polynomials. Now consider the $\pmod p$ reductions $\overline{f}, \overline{g}, \overline{h}$. Suppose for contradiction that $\overline{f}$ is irreductible over $\mathbb{Z}_p[x]$. Then since $\overline{f} = \overline{g}\overline{h}$ since reduction is a homomorphism, and since $\deg \overline{f} = \deg f$, we must have $\deg \overline{g} < \deg g$, and likewise for h. However, we have $\deg \overline{f} = \deg \overline{g} + \deg \overline{h}$, a contradiction. ■

Lecture 7: More More on Polynomials

Mon 10 Feb 2025 13:25

Here's a theorem proven by Eisenstein in 1850.

Theorem 1.5

Let $f(x) \in \mathbb{Z}[x]$. If there exists a prime p that does not divide a_n, but p divides a_i for $0 \leq i < n$ and p^2 does not divide a_0, then $f(x)$ is irreductable over $\mathbb{Q}$.

Proof. Recall that if f is irreducible over $\mathbb{Q}$, then it is irreducible over $\mathbb{Z}$ since $\mathbb{Z} \subseteq \mathbb{Q}$. Suppose $f = gh$ for $g, h \in \mathbb{Z}[x]$ where $1 \leq \deg g < n$, and same for h. Note that $p|a_0$, but p^2 does not divide a_0. We must have $p|b_0c_0$, and thus $p|b_0$ xor $p|c_0$. Since p does not divide $a_n = b_rc_s$, then p does not divide b_r, so there is a least integer t such that p does not divide b_t. We find that

$$a_t = b_tc_0 + \cdots + b_0c_t.$$

By assumption, p divides a_t, but this is impossible because p does not divide b_t and c_0. ■

This theorem allows us to manufacture many examples of irreductable polynomials of any degree. For example, $f(x) = x^5 - 9x^4 + 3x^2 - 12$ satisfies thee conditions for $p = 3$, so it is not factorable over $\mathbb{Q}$.

An important class of examples this criteria is used for is the **cyclotomic** polynomials. Define

$$\Phi_p(x) := \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1.$$

The roots of $x^p - 1$ are the p-th roots of unity, which corresponds to chopping up the circle group p times. It can be shown the pth roots of unity are ζ_p^i, for

$$\zeta_p = e^{i\frac{2\pi}{p}}.$$

Note that $\zeta_p^p = \zeta_p^0$. Moreover, each ζ_p^k is distinct for all k from 0 to $p-1$. The cyclotomic polynomials have all solutions ζ_p^k other than $x = 1$.

Corollary 1.4

The cyclotomic polynomials Φ_p are irreducible over $\mathbb{Q}$.

Proof. Recall the binomial theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

If p is prime, then

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}.$$

Since p is prime, if k is between $0, p$, then p cannot divide $k!$. Additionally, p does not divide $(p-k)!$.

Let

$$f(x) = \Phi_p(x+1) = \frac{(x+1)^p - 1}{(x+1) - 1} = \sum_{k=1}^{p} \binom{p}{k-1} x^{p-k}.$$

Every coefficient of $f(x)$ except that of x^{p-1} is divisible by p, since $p | p!$, and notably p^2 does not divide p choose $p-1$. By theorem 1.5, $f(x)$ is irreducible over $\mathbb{Q}$. Certaintly, if $\Phi_p(x) = g(x)h(x)$, then $f(x) = g(x+1)h(x+1)$, so we must have that Φ_p is irreducible. ■

Now let's discuss factorization in $\mathbb{F}[x]$. As we will see, $\mathbb{F}[x]$ has a division algorithm similar to that in $\mathbb{Z}$.

In the naturals, one of the main properties of primes is not that their only divisors are 1 and themselves, but that if $p|rs$ then $p|r$ or $p|s$. In $\mathbb{F}[x]$, irreducible polynomials play this role.

Theorem 1.6

Let $p(x) \in \mathbb{F}[x]$ be irreducible. If $p(x)$ divides $r(x)s(x)$, then $p(x)|r(x)$ or $p(x)|s(x)$.

Proof. later ■

Corollary 1.5

Let $p(x) \in \mathbb{F}[x]$ be irreducible, and let it divide a product $r_1(x) \cdots r_n(x) \in \mathbb{F}[x]$. Then $p(x)$ divide $r_i(x)$ for at least one $i \in \{1, \ldots, n\}$.

Proof. Follows immediately from theorem 1.6 by induction. ■

Theorem 1.7

Let $\mathbb{F}$ be a field. Then for every $f(x) \in \mathbb{F}[x] \setminus \mathbb{F}$, there exists a factorization in $\mathbb{F}[x]$ into irreducibles, where the irreducibles are unique up to order and constant factors in $\mathbb{F}$. That is, if

$$f(x) = p_1 \cdots p_r = q_1 \cdots q_s,$$

for p_i, q_i irreducibles, then $r = s$ and $p_i = u_i q_i$ for $u_i \in \mathbb{F}$ for all i.

We will prove this statement more generally later, as a statement about divisibility in integral domains.

Chapter 2

Ideals and Quotients

Lecture 8: Ideals and Quotient Rings

Wed 12 Feb 2025 13:27

Definition 2.1: Ideal

A subring I of a ring R is a (two-sided) ideal of R if for all $r \in R$ and $a \in I$, $ra, ar \in I$. Equivalently, for all $r \in R$, it follows that $rI, Ir \subseteq I$.

Proposition 2.1

A nonempty subset $I \subseteq R$ is an ideal of R if

- $a - b \in I$ for all $a, b \in I$;
- $ra, ar \in I$ for all $a \in I$ and $r \in R$.

Proof. trivial. ■

Definition 2.2: Principal Ideal

Let R be a commutative ring, and let $a \in R$. Define

$$\langle a \rangle \coloneqq \{ra \mid r \in R\}.$$

We call $\langle a \rangle$ the *principal ideal* generated by a.

Proposition 2.2

Principal ideals are ideals.

Proof. Let $ra, sa \in \langle a \rangle$. Then $ra - sa = (r - s)a \in \langle a \rangle$ since $r - s \in R$. By definition, $ra \in \langle a \rangle$ for all $r \in R$, so by proposition 2.1, $\langle a \rangle$ is an ideal. ■

Consider the very important example of $\mathbb{R}[x]$. Consider I as the set of polynomials with a constant term equal to 0. That is, $I = \langle x \rangle$. We can thus find that $I = x\mathbb{R}[x]$ (we can factor out x from every element of $\langle x \rangle$).

We can also consider ideals with multiple generators.

Definition 2.3: Generated Ideal

Let $X \subseteq R$ be a set. Define

$$\langle X \rangle := \left\{ \sum_{i=0}^{k} r_i x_i \mid \{x_i\}_{i=0}^{k} \subseteq X, \{r_i\}_{i=0}^{k} \subseteq R \right\}.$$

That is, $\langle X \rangle$ is the collection of all R-linear combinations.

The set $\langle X \rangle$ is trivially shown to be an ideal. We can define $R[x,y] := R[x][y]$. I don't know why we're talking about this. Now consider cosets. We already know a lot about cosets from group theory; for example, $a - b \in I$ if and only if $a + I = b + I$. Let's apply this to the example $\mathbb{R}[x]$ together with the ideal $\langle x \rangle$.

Suppose that $f(x), g(x) \in \mathbb{R}[x]$ such that $f(x) + \langle x \rangle = g(x) + \langle x \rangle$. Note that we must have $f(x) - g(x) \in I$, and thus $f(0) - g(0) = 0$. Equivalently, we have that x divides $f(x) - g(x)$, so if we define $h(x) = f(x) - g(x)$, then $h(x) = xq(x)$ for some $q \in \mathbb{R}[x]$, and thus $h(0) = 0$. Two ways of arriving at this fact. This implies that $f(0) = g(0)$, and thus f, g have the same constant terms, as $f(0) = a_0$ clearly. We thus find that $f(x) + \langle x \rangle = g(x) + \langle x \rangle$ if and only if f, g have the same constant term.

Lecture 9: Quotient Rings

Fri 14 Feb 2025 13:32

Definition 2.4: Quotient Ring

Let R be a ring with a two-sided ideal $I \subseteq R$. Then the collection of cosets R/I, together with the operations of addition and multiplication

$$(r + I) + (s + I) = (r + s) + I,$$

$$(r + I)(s + I) = rs + I$$

is called the quotient ring of R modulo I.

Lecture 10: First Isomorphism Theorem and Friends

Tue 18 Feb 2025 13:25

What is the structure of a quotient ring? We keep all facts from group theory, such as

$$x + I = y + I \implies x + z + I = y + z + I, \qquad x + I = I \iff x \in I,$$

$$x - y \in I \iff x + I = y + I.$$

We can do some interesting things with these rules. For example, if for some reason we want $2 = i$ in $\mathbb{Z}[i]$, then we can write $2 - i = 0$, and then consider the ring

$$\frac{\mathbb{Z}[i]}{\langle x - i \rangle}.$$

However, we can in fact explore this ring further, and using elementary arithmatic show the only distinct cosets are $r + I$ for $r \in \{0, 1, 2, 3, 4\}$. We in fact have

$$\frac{\mathbb{Z}[i]}{\langle 2 - i \rangle} \cong \mathbb{Z}_5.$$

This is a field, and thus $\langle 2 - i \rangle$ is maximal.

Similarly to group theory, ideals are preciely the kernels of ring homomorphisms.

Definition 2.5: Kernel

Let $\phi : R \to S$ be a ring homomorphism. Then $\ker \phi$ is defined as the set

$$\ker \phi \coloneqq \{r \in R \mid \phi(r) = 0\}.$$

Proposition 2.3

Let $\phi : R \to S$ be a ring homomorphism. Then

- $\phi(0_R) = 0_S$;
- $\phi(-r) = -\phi(r)$;
- If $R' \subseteq R$ is a subring, then $\phi(R') \subseteq S$ is a subring;
- If $S' \subseteq S$ is a subring, then $\phi^{-1}(S') \subseteq R$ is a subring;
- If R has unity 1_R, then $\phi(R)$ has unity $\phi(1_R)$ (but not necessarily all of S);
- if I is an ideal of R, then $\phi(I)$ is an ideal of $\phi(R)$;
- If J is an ideal of S, then $\phi^{-1}(J)$ is an ideal of R.

Proposition 2.4

A homomorphism $\phi : R \to S$ is injective if and only if $\ker \phi$ is trivial.

Proof. Let $\phi : R \to S$ have a trivial kernel. Let $r, s \in R$ such that $\phi(r) = \phi(s)$. It follows that $\phi(r - s) = 0_S$, and thus $r - s \in \ker \phi$. It follows that $r - s = 0_R$, and thus $r = s$.

Let $\phi : R \to S$ be injective. Note that $0_R \mapsto 0_S$, so $0_R \in \ker \phi$. It follows that for all $x \in \ker \phi$, $k \mapsto 0_S$ implies $x = 0_R$ because ϕ is injective. ■

Note that since $\{0\}$ is an ideal in all rings, and the kernel is just $\phi^{-1}\{0\}$, we have that $\ker \phi$ is an ideal.

Theorem 2.1 (First Isomorphism Theorem)

Let $\phi : R \to S$ be a ring homomorphism. Then

$$\frac{R}{\ker \phi} \cong \phi(R).$$

Proof. Let $\phi : R \to S$, and let $\pi : R \to R/\ker \phi$ be the canonical projection. Since $R/\ker \phi$ is a quotient, there exists a unique injection $\psi : R/\ker \phi \to S$, given by $r + I \mapsto r$. Restricting the codomain to $\phi(R)$ makes ψ manifestly bijective, and thus an isomorphism. ■

Corollaries include that if ϕ is injective, then $\phi(R) \cong R$, and if ϕ is surjective, then $R/\ker \phi \cong S$.

Lecture 11: Ideals and Isomorphism Theorems

Wed 19 Feb 2025 13:25

Proposition 2.5

Let I, J be ideals of R. Then $I + J$ and $I \cap J$ are ideals of R.

Proof. ive alr done this twice wtf idc ■

Theorem 2.2 (Second Isomorphism Theorem)

Let I, J be ideals of a ring R. Then
$$\frac{I+J}{J} \cong \frac{I}{I \cap J}.$$

Proof. owo ■

Theorem 2.3 (Third Isomorphism Theorem)

Let R be a ring with ideals I, J such that $I \subseteq J$. Then J/I is an ideal of R/I, and
$$\frac{R/I}{J/I} \cong \frac{R}{J}.$$

Proof. Follows easily from the first isomorphism theorem. ■

2.1 Prime and Maximal Ideals

We spent all of class motivating it.

Definition 2.6: Prime/Maximal Ideals

Let R be a ring, and I a proper ideal of R. Then I is prime if and only if
$$ab \in I \implies a \in I \vee b \in I.$$
A maximal ideal is an ideal I such that if there exists an ideal J with $I \subseteq J$, then $J = R$.

Chapter 3

Divisibility

Lecture 12: Divisibility in Integral Domains

Mon 24 Feb 2025 13:25

Definition 3.1

Let D be an integral domain.

- Elements $a, b \in D$ are *associates* if $a = ub$ for some $u \in D$.
- A nonzero $a \in D$ is called *prime* if a is not a unit, and $a|bc$ implies $a|b$ or $a|c$.
- A nonzero element $a \in D$ is *irreducible* if a is not a unit, and whemever $a = bx$ for $b, c \in D$, it follows that that b, c are units.

It's simple to show in a domain that prime $\implies$ irreducible. However, is the converse true? It turns out it is *not*, and we can show this via counterexample. The counterexample took half of class so I zoned out.

Definition 3.2: PID

An integral domain R is called a Principle Ideal Domain (PID) if all ideals in R are principal; that is, if I is an ideal of R, then there exists $r \in R$ such that $I = \langle r \rangle$.

Lecture 13: New Stuff

Wed 26 Feb 2025 13:26

Theorem 3.1

In a PID, every irreducible element is prime.

Proof. Let $a \in D$ be irreducible in a PID. Suppose that $a|bc$, for $b, c \in D$. Let $I = \langle a, b \rangle$, and since D is a PID, we have that $I = \langle d \rangle$ for some $d \in D$. Since $a \in I$, we have $a = dr$ for some $r \in D$. Since a is irreducible, either d or r is a unit. If d is a unit then $I = D$ trivially, and thus $1 = ax + by$ for some $x, y \in D$, and thus $c = acx + bcy$. Since $a|bc$ and $a|ac$, it follows that $a|c$. If r is instead the unit, then $\langle a \rangle = \langle d \rangle$, since any $ka = kdr$ and $kd = kar^{-1}$. Since $b \in I$, there is a $t \in D$ such that $at = b$, and thus $a|b$. ∎

Corollary 3.1

If D is a PID, then $p \in D$ is prime if and only if it is irreducible.

Corollary 3.2

If there exists an irreducible $a \in D$ of a domain that is not prime, then D is not a PID.

Definition 3.3: Unique Factorization Domain

An integral domain D is a unique factorization domain (UFD) iff

- Every nonzero nonunit of D is equivalent to a product of irreducibles in D.
- This factorization is unique up to the order of the irreducibles and the associates.

The integers are a UFD by the fundamental theorem of arithmetic.

Lemma 3.1

If D is a domain, then for all units $u \in D^*$, we have $\langle u \rangle = D$.

Proof.

$$uu^{-1} = 1_D.$$

■

Lemma 3.2

Let D be a domain. Then $a = bc$ for b, c not units if and only if $\langle a \rangle \subsetneq \langle b \rangle$.

Proof. Since a is a multiple of b, we have $\langle a \rangle \subseteq \langle b \rangle$. The containment is proper if c is a non-unit trivially. ■

This lemma implies we can have a sequence of proper inclusions

$$\langle a \rangle \subsetneq \langle b \rangle \subsetneq \cdots$$

which corresponds to factorization. The question is whether this chain terminates finitely.

Lecture 14: Ascending Chains

Fri 28 Feb 2025 13:34

Definition 3.4: ACC

Let R be a cmmutative ring. We say the ascending chain condition holds if any chain of ideals

$$I_1 \subseteq I_2 \subseteq \ldots$$

there exists r such that $I_{r-1} \subseteq I_r = I_{r+1} = \ldots$. That is, the chain stabilizes.

Lemma 3.3

Every PID satisfies ACC.

Proof. Let $I_\bullet$ be a chain in the PID D. Let

$$J = \bigcup_{\alpha \in \mathbb{N}} I_\alpha.$$

We can show J is an ideal. If $x, y \in J$, then $x \in I_m$, $y \in I_n$. WLOG, let $m \leq n$. Thus, $x, y \in I_n$, and thus $x - y \in I_n$. Note that for any $x \in J$, we have $x \in I_k$, and $ax \in I_k$ for all $a \in R$. Therefore, J is an ideal. Since J is an ideal, $J = \langle z \rangle$ for some $z \in D$. It follows that $z \in I_r$ for some r, so $\langle z \rangle \subseteq I_r$. Therefore, there exists an ideal in $I_\bullet$ which is the upper bound. ■

Theorem 3.2

Let D be a PID. Then D is a UFD. In particular, the category of PIDs is a subcategory of the category of UFDs.

Proof. First, we will show every nonzero nonunit is divisible by at least one irreducible. Let $a \in D$. If a is irreducible, then $a|a$. If it's not irreducible, then it's reducible, and thus there exist nonzero nonunits such that $a = x_1 y_1$. It follows that $\langle a \rangle \subsetneq \langle x_1 \rangle$. If x_1 is not irreducible, then we have $\langle a \rangle \subsetneq \langle x_1 \rangle \subsetneq \langle x_2 \rangle$ for some x_2. By induction, we obtain a chain $x_\bullet$. Since D is a PID, by lemma **??** there exists r such that $\langle x_r \rangle$ is maximal in this chain. Therefore, x_r is irreducible, and divides a. Moreover, a is the product of irreducibles.

Now let $p_1 \ldots p_n$ and $q_1 \ldots q_m$ be irreducible factorizations of a. Since p_1 is a factor of a,

$$p_1 | q_1 \ldots q_m.$$

Since p_1 is a prime by theorem 3.1, p_1 divides some q_i. It follows that q_i is not irreducible unless $p_1 = q_1$, since they are not units. It follows fairly trivially that D is a UFD. ■

Corollary 3.3

Let F be a field. Then $F[x]$ is a UFD.

Proof. Since $F[x]$ is a PID, this follows from theorem 3.2. ■

But is $\mathbb{Z}[x]$ a UFD? It's not a PID, but $\mathbb{Z}$ is a PID, so it is a UFD. We then have the following.

Theorem 3.3

Let R be a UFD. Then $R[x]$ is a UFD. By extension, $R[A]$ is a UFD for some (at least finite, maybe it can be infinite) set A.

Proof. we didnt do it lol ■

Lecture 15: Euclidean Domains

Mon 03 Mar 2025 13:28

Definition 3.5: Euclidean Domain

An integral domain D is called a Euclidean Domain if there exists a function $d : D^* \to \mathbb{N}$ such that

- $d(a) \leq d(ab)$ for all $a, b \in D^*$;
- If $a, b \in D$ with $b \neq 0$, then there exist $q, r \in D$ such that $a = qb + r$ where either $r = 0$ or $d(r) < d(b)$.

A Euclidean domain is kind of between fields and PIDs, since there is a notion of division given by the division algorithm.

For example, $F[x]$ with d the degree function $\deg f$. $\mathbb{Z}$ is also trivially an example.

Theorem 3.4

Every Euclidean Domain is a PID.

Proof. we just didn't do it. ■

Chapter 4

Field Extensions

Lecture 16: Extension Fields

Wed 05 Mar 2025 13:28

Definition 4.1: Field Extension

A field E is an extension field of F if $F \subseteq E$ and the inclusion $F \hookrightarrow E$ is a homomorphism.

Theorem 4.1 (Fundamental Theorem of Field Theory)

Let F be a field and let $f(x) \in F[x]$. Then there is an extension field E of F containing a root of $f(x)$.

Proof. GIven $f(x)$, let $p(x)$ be an irreducible factor of $f(x)$, which exists because $F[x]$ is a UFD. Let

$$E = \frac{F[x]}{\langle p(x) \rangle}.$$

This is a field because $p(x)$ is irreducible, and thus its ideal is maximal. We can view E as an extension field of F by the inclusion $F \hookrightarrow F[x]$ composed with the projection $F[x] \to E$.

Let $\alpha = x + \langle p(x) \rangle$, and consider $p(\alpha)$. This simplifies to $p(x) + \langle p(x) \rangle$, and thus $\langle p(x) \rangle$, and we have that E contains a root of $p(x)$, which is a factor of $f(x)$. ■